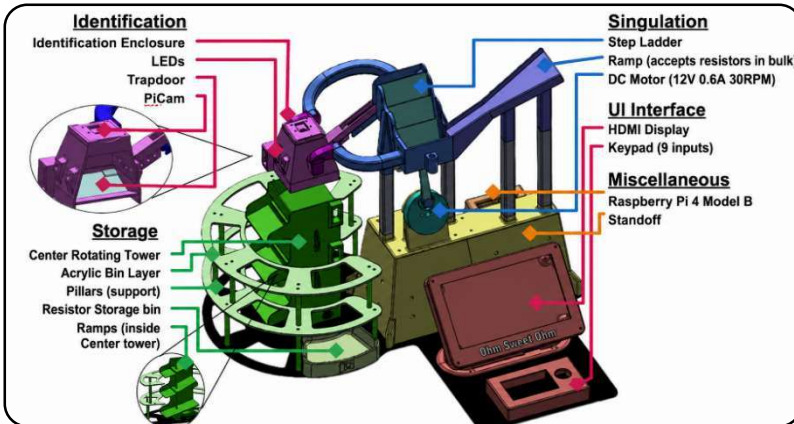




Automated Resistor Sorter



Results

- Achieved **95% overall sorting accuracy** across **14 resistor values** at **~11 resistors/min** throughput with **fully autonomous operation**
- Completed **15+** prototype iterations and subsystem tests to validate design decisions
- Engineered a **low-friction** singulation mechanism with **optimized tolerances** to achieve **smooth, quiet, and jam-free separation**
- Developed **scalable storage architecture** capable of holding **1,000+ resistors** with additional capacity for future expansion
- Designed **modular subsystems** enabling **rapid assembly, maintenance, and future upgrades**
- Final design accommodated **15–20°** resistor lead bends while maintaining reliable sorting performance and achieved an estimated manufacturing cost of **~\$220/unit** using market-price components

Informational Video Demo:

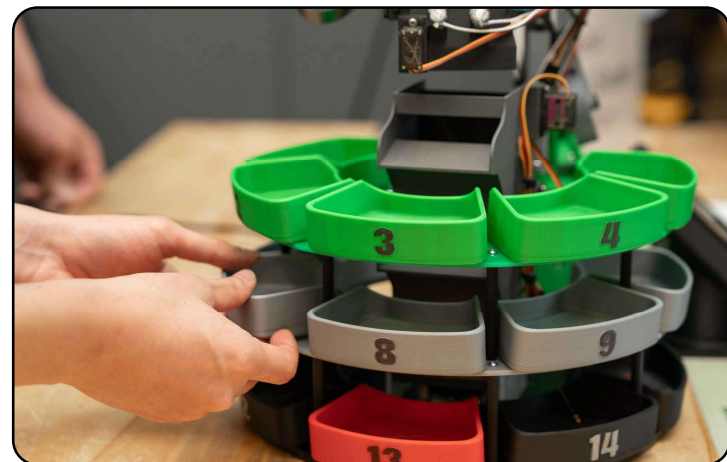
<https://www.youtube.com/watch?v=gMguH-DOG48>

What?

- The **Automated Resistor Sorter** is an automated machine that **singulates, identifies, and sorts through-hole resistors**
- Developed to **reduce manual sorting time** while improving consistency and scalability for prototyping environments

Challenge

- Conducted **customer discovery, competitor benchmarking, and engineering requirements analysis** to establish design specifications
- Generated and evaluated multiple design concepts using a **weighted Pugh Matrix**
- Refined additive manufacturing settings** via **iterative testing** to reduce **surface friction**, resulting in **smooth, repeatable operation**
- Developed and tested the final singulation mechanism, serving as the **primary contributor** for its **concept, CAD, and validation**



Demonstration of Product